Carbohydrates

Carbohydrates (sugars, saccharides or glucides) are the most abundant organic molecules in nature. Many simple carbohydrates contain C, H and O in the approximate stoichiometric ratio $(CH_2O)_n$, which expressed as $C_n(H_2O)_n$ explains the name carbohydrate. However, many derivatives do not retain this formula and may contain N, P or S. They vary greatly in size (from less than 100 Da to millions of Da), because they can exist in the form of free monomers (called **monosaccharides**) or as chains with a few or several covalently linked monosaccharides (**oligosaccharides** or **polysaccharides**, respectively). Besides, carbohydrates can bind to proteins and lipids, giving rise to **glycoconjugates**.

Biological functions of carbohydrates

- **Energy source:** together with fats, carbohydrates are the main source of energy from our diet. For example, under normal physiological conditions, glucose is a major energy source for the brain, whereas cells such as red blood cells depend entirely on glucose for energy.
- **Energy storage:** carbohydrates are energy-rich compounds and can be mobilized quickly, which is why they are used for energy storage (as glycogen in animals and starch in plants).
- **Structural function:** insoluble polymers of carbohydrates act as structural elements in cell walls, cell membranes and the connective tissue. Examples are cellulose (cell wall of plants), chitin (exoskeleton of arthropods), or glycoproteins and glycolipids in membranes and the extracellular matrix.
- **Precursor function:** many carbohydrates are biogenic compounds, that is, they are used as precursors for the synthesis of many other biomolecules. For example, ribose is a component of nucleic acids, coenzymes, etc.
- **Regulatory function**: carbohydrates can participate in intracellular signaling (for example, inositol trisphosphate), glycoproteins and glycolipids in cell membranes are important in cell-cell signaling and recognition, etc.

Classification of carbohydrates

There are three main classes of carbohydrates depending on the number of monomers:

1- **Monosaccharides**: they can be simple or derived.

2- **Oligosaccharides**: short chains containing a small number of monosaccharide units (around 2-20). Disaccharides are the simplest and most common examples. Larger oligosaccharides are generally conjugated with lipids or proteins forming glycolipids or glycoproteins, respectively.

3- **Polysaccharides**: they originate from the union of hundreds, or even thousands, of monosaccharides. Depending on their function, they are divided into **structural polysaccharides** and **reserve polysaccharides**. Depending on the structure of the chain, they can be **linear** or

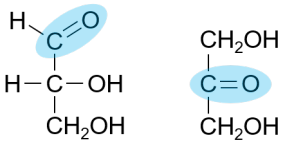
branched. Depending on their components, they can be **homopolysaccharides**, with only one type of monosaccharide, or **heteropolysaccharides**, with more than one type of monosaccharide.

Nomenclature

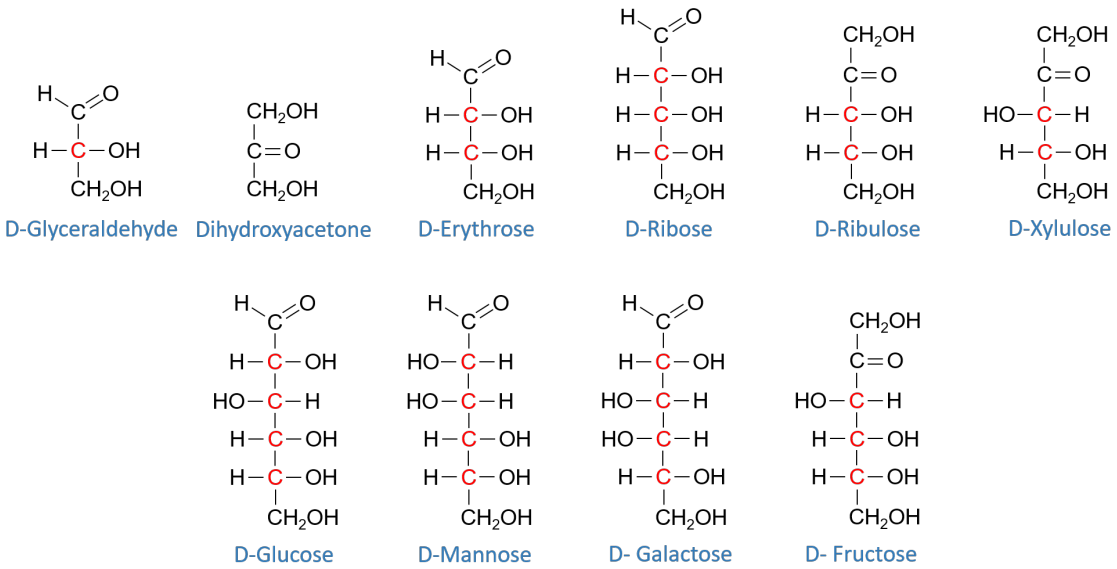
There is no systematic nomenclature for the vast majority of carbohydrates. Rather, a common name is used for each of them. In the case of monosaccharides and disaccharides the common names typically end in **-ose**.

Monosaccharides

Monosaccharides are either **aldehydes or ketones with two or more hydroxyl groups**. In their open-chain form, monosaccharides consist of a carbon chain containing several hydroxyl groups and one carbonyl group. **Aldoses** contain an aldehyde group, whereas **ketoses** contain a ketone group. In other words, monosaccharides are polyhydroxyaldehydes (aldoses) or polyhydroxyketones (ketoses). They are **soluble in water** and insoluble in non-polar solvents. Many simple monosaccharides have a sweet taste.



Most biologically relevant monosaccharides contain three to seven carbon atoms. The most common monosaccharides are trioses, tetroses, pentoses and hexoses. Shown below are some representative monosaccharides.



All monosaccharides, except for dihydroxyacetone, have at least one **asymmetric carbon** (labeled in red in the image above). The four groups of atoms attached to the asymmetric carbon can be arranged in space in different ways. The different spatial arrangements give rise to **stereoisomers**. **The great majority of monosaccharides found in nature are D-stereoisomers**. In a Fischer projection, D-monosaccharides have the hydroxyl group on the chiral carbon farthest from the carbonyl group positioned on the right.

Watch this video to learn about the different stereoisomers present in monosaccharides

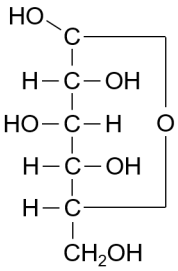
Reducing power of monosaccharides

Most monosaccharides behave as reducing sugars. The carbonyl carbon in aldoses can be oxidized to carboxylic acid by relatively mild oxidizing agents. Ketoses can also behave as reducing sugars under appropriate conditions because they can isomerize to aldoses. This has practical implications. The concentration of glucose in blood or urine can be determined by an enzyme-catalyzed reaction based on the reducing capacity of glucose:

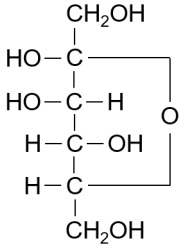
$$\text{D-Glucose} + O_2 \xrightarrow{\textit{Glucose oxidase}} \text{D-Glucono-}\delta\text{-lactone} + H_2O_2$$

Cyclization of monosaccharides

Although we frequently represent the structures of monosaccharides as straight-chain molecules, in fact, **monosaccharides containing five or more carbon atoms predominantly exist as cyclic structures in aqueous solution**. Monosaccharide rings are generated by the reaction of the carbonyl group with a hydroxyl group of the monosaccharide itself, forming a hemiacetal (in the case of aldoses) or a hemiketal (in the case of ketoses). Six-membered rings are known as **pyranoses** and five-membered rings as **furanoses**. The carbon involved in the cyclization of the monosaccharide is known as the **anomeric carbon** and is an asymmetric carbon. The two stereoisomers of a cyclic monosaccharide are known as anomers (α and β). In solution, ring opening and reclosure can result in the interconversion between anomers. This process is called **mutarotation**.

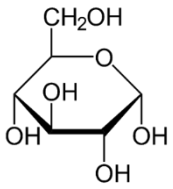


D-Glucopyranose

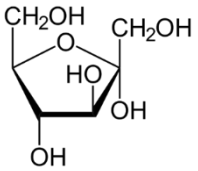


D-Fructofuranose

The **Haworth projection** is the most common way of representing the cyclic structure of monosaccharides.



α-D-Glucopyranose



α-D-Fructofuranose



Watch this video to learn how to draw the Haworth projection from the Fischer projection of a monosaccharide

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Some hints about the Haworth projection formula:

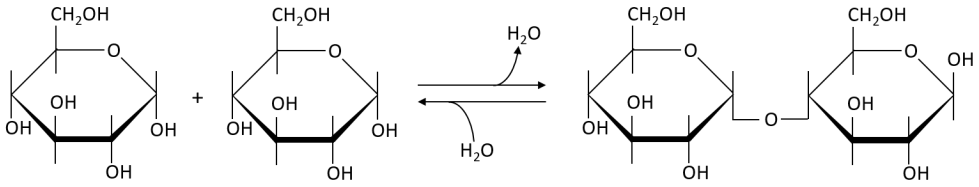
- *The ring is drawn as if it were approximately perpendicular to the plane of the page. Bonds closest to the viewer are sometimes drawn thicker to give a sense of perspective)*
- *The oxygen atom occupies the upper right-hand corner (in pyranoses) or the upper center (in furanoses)*
- *The substituents that are to the right in the Fischer projection are down in the Haworth projection. The substituents that are to the left in the Fischer projection are up in the Haworth projection*
- *In D stereoisomers, the terminal -CH_2OH group is located above the plane of the ring*

Monosaccharide derivatives

In many cases, the hydroxyl groups on monosaccharides are replaced by other substituents, or the carbon atoms are oxidized to carboxylic acids or reduced to alcohols. These and many other possible modifications give rise to derivatives of monosaccharides with multiple functions in the body (deoxyribose is part of DNA, carbohydrate metabolites are frequently phosphorylated, etc.). Many derivatives are found in oligosaccharides that are part of glycolipids and glycoproteins, or in polysaccharides.

Disaccharides

A disaccharide is formed by **two monosaccharides bound through an O-glycosidic bond**. An O-glycosidic bond is a covalent bond formed between the anomeric carbon of one monosaccharide and an oxygen atom from a hydroxyl group of another molecule (another monosaccharide, in the case of disaccharides). The formation of this bond is a reaction of **condensation**; whereas its breakdown is a reaction of **hydrolysis**.



Classification of disaccharides

Disaccharides are grouped into:

- **Reducing disaccharides**: only one anomeric carbon participates in the glycosidic bond. The other anomeric carbon remains free and retains its reducing capacity.
- **Non-reducing disaccharides**: both anomeric carbons participate in the glycosidic bond. Therefore, there is no free anomeric carbon and the disaccharide has no reducing power.

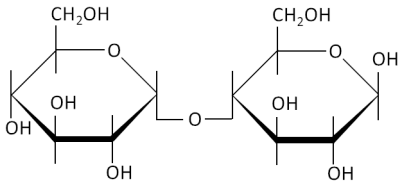
Nomenclature

Disaccharides are named by a common name or by a name that refers to the monosaccharides and the linkage type, following this sequence:

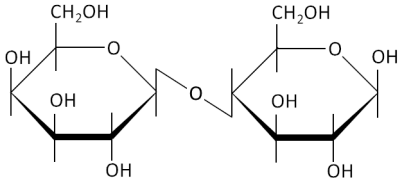
- The monosaccharide whose anomeric carbon forms the glycosidic bond, ending in *-osyl*.
- The configuration (α or β) of the anomeric carbon and the positions of the carbon atoms involved in the glycosidic bond.
- The second monosaccharide ending in *-ose* if its anomeric carbon remains free. If its anomeric carbon also participates in the glycosidic bond, it ends in *-oside*.

The configuration of the anomeric carbon is of special importance because there is not possibility of interconversion with the other anomer.

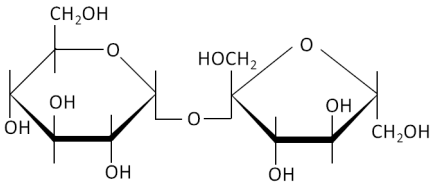
Shown below are some representative disaccharides.



Maltose
α-D-glucopyranosyl-(1→4)-D-glucopyranose



Lactose
β-D-galactopyranosyl-(1→4)-D-glucopyranose



Sucrose
α-D-glucopyranosyl-(1→2)-β-D-fructofuranose

Maltose is formed by the partial hydrolysis of starch and glycogen in the diet, and consists of **two glucose residues linked by an α 1→4 bond**. A subsequent hydrolysis catalyzed by enzymes such as *maltase* results in the formation of glucose, which can then be absorbed by intestinal epithelial cells

Lactose is present in the milk of humans and other mammals. Lactose is a reducing sugar composed of **one molecule of D-galactose and one molecule of D-glucose joined by a β 1→4 glycosidic bond. Lactose intolerance** is caused by an insufficient production of ***lactase***, the enzyme that catalyzes the hydrolysis of lactose into glucose and galactose in the small intestine. Individuals with this deficiency cannot digest the lactose found in milk. A more serious problem is the genetic disease **galactosemia**, an inherited disorder caused by defects in enzymes involved in galactose metabolism.

Sucrose consists of **glucose and fructose linked through an α 1→2 glycosidic bond** involving the anomeric carbon of both monosaccharides. Sucrose is commonly known as table sugar, and is obtained from sugar cane or sugar beets. Sucrose contains no free anomeric carbons; therefore, it is a non-reducing sugar.

Polysaccharides

Polysaccharides, also called glycans, are **the most abundant form of carbohydrates in nature**. They are very large **polymers composed of tens to thousands of monosaccharide units linked together by O-glycosidic bonds**. Because they contain only one potentially reducing end per very large molecule, polysaccharides show negligible reducing activity. Their solubility varies widely depending on their structure. They are generally not sweet and do not show appreciable mutarotation.

Most of them have names assigned without any systematic rule. The identity of each polysaccharide is determined by:

1- The nature of its monomeric units. According to this characteristic, there are:

- **Homopolysaccharides**: by repetition of a single monosaccharide (glucans, mannans, fructans, etc.).
- **Heteropolysaccharides**: two or more types of monosaccharides in the polymer.

2- The types of bonds that are formed between the units: both the bond position and the configuration of the anomeric carbons are important.

3- The **length** of the chain: unlike other biopolymers, such as proteins, polysaccharides do not have a defined number of residues, although their length varies within some limits.

4- The shape of the chain: **linear** or **branched**. The degree of branching can also be different.

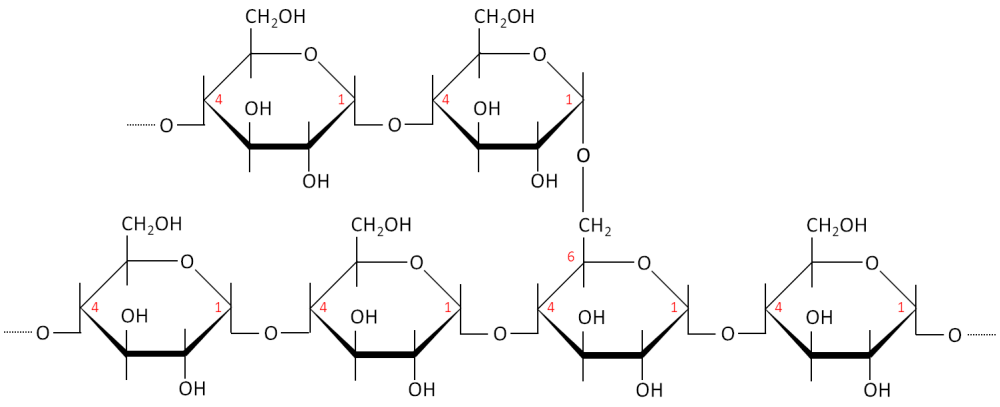
Polysaccharides can be classified by their function in:

- **Storage polysaccharides**: they serve as energy reserves.
- **Structural polysaccharides**: they provide mechanical support to biological structures.

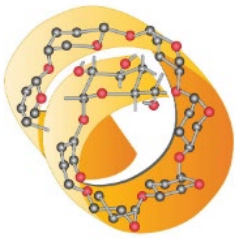
Storage polysaccharides

Storage polysaccharides are homopolysaccharides. They store monosaccharides that can be used as an energy source to carry out biological processes. The most important reserve polysaccharides are glycogen in animal cells and starch in plant cells. Both are found in the cytoplasm forming granules.

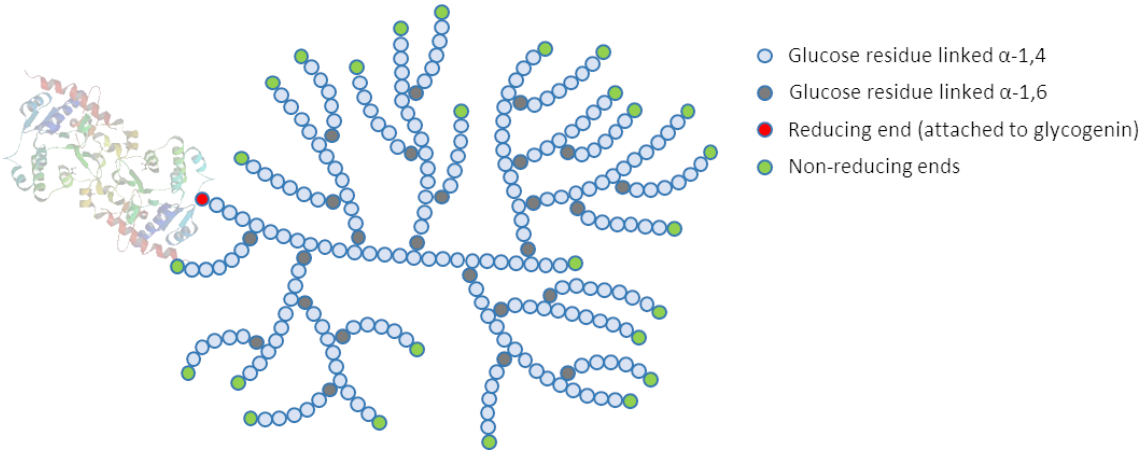
Glycogen consists of **linear chains of glucose residues linked by α 1→4 bonds with branches in α 1→6 positions**. Branches appear approximately every 10 residues and give rise to a "tree" structure. Practically all mammalian cells contain glycogen, but it is especially abundant in the liver (4–8% by weight of tissue) and in skeletal muscle cells (0.5–1%). Despite the lower proportion, about 70% of the total glycogen in the body is stored in muscle cells.



Regarding the spatial disposition, glucose molecules acquire a **helical arrangement** due to the bond angle that the α configuration of the anomeric carbon imposes. The helical structure is stabilized by hydrogen bonds between the -OH groups of glucose residues. This disposition results in a very compact polymer.



Because of its extensive branching, glycogen has many non-reducing ends per reducing end. This is biologically interesting since, when glycogen is used as a source of energy, **glucose units are enzymatically removed one by one from the abundant non-reducing ends of glycogen**. The reducing end of glycogen is attached to **glycogenin**, the protein from which glycogen synthesis begins.



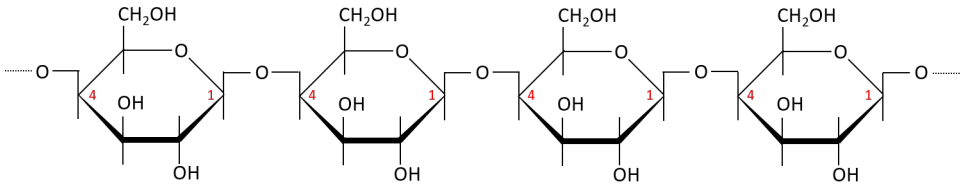
Starch is the most important carbohydrate in human nutrition, accounting for more than 50% of our carbohydrate intake. Starch is a mixture of two polymers: amylose (15-20%) and amylopectin (80-85%). Amylose is a linear polymer of glucose residues joined by α1→4 linkages. Amylopectin is structurally similar to glycogen but is less highly branched, with longer α1→4-linked chains between α1→6 branch points.

Humans have enzymes that catalyze the degradation of the starch in the diet. **α-Amylases**, secreted by the salivary glands and the pancreas, catalyze the hydrolysis of α1→4 bonds, giving rise to fragments of 2-9 glucoses, maltose (Glc α1→4 Glc), maltotriose (trisaccharide) and α-limit dextrins (branched oligosaccharide). These fragments are subsequently broken down to glucose units by other amylases and hydrolases.

Structural polysaccharides

Cellulose is the most abundant organic compound in the biosphere, accounting for around 50% of all the carbon found in organic molecules. Cellulose is found in plants, where it provides structural support to the cell wall.

Cellulose is a **linear polymer of β1→4 linked glucoses**. Chains are long: 10,000-15,000 residues.

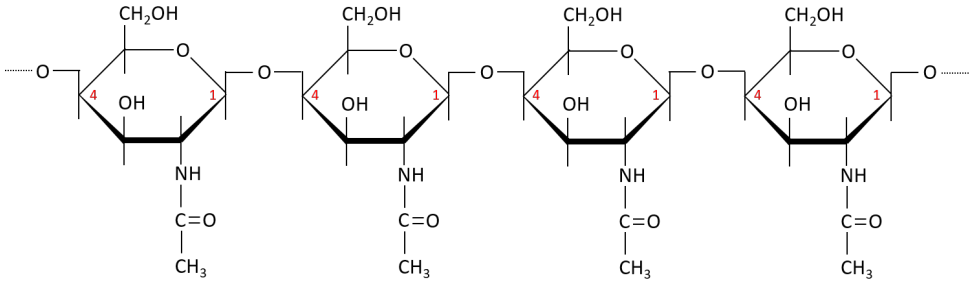


The β1→4 glycosidic bonds produce **extended, linear chains**, in contrast to the helical structures favored by α1→4-linked glucose polymers. Cellulose chains are stabilized by hydrogen bonds between neighboring glucoses. This gives rise to linear structures that are associated with each other by interchain hydrogen bonds. As a result, cellulose is packed into fibers with great resistance to tension. The large number of intramolecular hydrogen bonds also limits the ability to form hydrogen bonds with water. Cellulose fibers in turn associate with other polysaccharides forming a complex cross-linked structure.

Most animals, including humans, cannot use cellulose as a source of glucose. Humans lack enzymes capable of catalyzing the hydrolysis of β1→4 glycosidic bonds in cellulose. Still, cellulose is important in our diet because it contributes to dietary fiber.

Certain microorganisms secrete cellulases and can digest cellulose. The presence of these microorganisms in the digestive tract of ruminants allows these animals to use cellulose for energy. Termites also have protozoa in their intestines capable of digesting cellulose. Many fungi also produce cellulases, which allow them to use wood as a glucose source.

Chitin is another example of structural homopolysaccharide. Chitin is present in the cell wall of fungi and the exoskeleton of arthropods (e.g. crustaceans, arachnids and insects). Chitin is the second most abundant organic molecule in nature. Chitin is composed of units of **N-acetyl-glucosamine with β 1→4 linkages**. This configuration of the glycosidic bond results in the formation of extensive fibers that cannot be digested by most animals. Chitin forms a matrix on which mineralization also occurs.

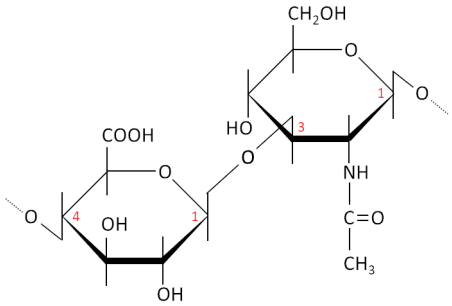


Peptidoglycan is a rigid component of bacterial cell walls. It is a **linear heteropolysaccharide of N-acetylglucosamine (GlcNAc) and N-acetylmuramic acid (MurNAc) linked by β 1⟶4 bonds**. The linear polymers are cross-linked by short peptides providing rigidity to the cell wall. ***Lysozyme***, an enzyme present in our tears, catalyzes the breakdown of the glycosidic bond between GlcNAc and MurNAc, destroying bacterial cells. **Penicillin** and related antibiotics act by inhibiting enzymes responsible for forming the peptide cross-links of peptidoglycan.

The extracellular matrix of our cells is rich in heteropolysaccharides that, together with fibrous proteins, provide strength and resistance to tissues. These heteropolysaccharides are known as **glycosaminoglycans**, which are **linear polysaccharides composed of repeating disaccharide units linked by β bonds**. Most glycosaminoglycans contain a uronic acid and an amino sugar. Their negatively charged carboxylate and sulfate groups strongly attract water and counterions, generating highly hydrated gels. This raises the viscosity of the medium and gives it an elastic consistency providing a flexible support for the extracellular matrix.

Some examples of glycosaminoglycans are **hyaluronic acid**, **chondroitin sulfate**, **keratan sulfate**, **heparan sulfate** and **heparin**.

Hyaluronic acid, or hyaluronan, is a characteristic element of connective tissue that acts as a major component of the extracellular matrix. It is found in large quantities in the vitreous humor and in the synovial fluid of the joints, cartilage, and tendons. It is a viscous, high molecular weight polysaccharide, that acts as a lubricating agent. It is made up of disaccharides linked by β 1⟶4 bonds. The disaccharide is glucuronic acid linked to N-acetylglucosamine via β 1⟶3 bonds. Hyaluronan is the only glycosaminoglycan that is not sulfated.



Hyaluronidase is an enzyme secreted by pathogenic bacteria that catalyzes the hydrolysis of hyaluronic acid, reducing the viscosity of the extracellular medium, and making tissues more susceptible to bacterial invasion.

Heparin has an unusually highly sulfated structure containing both α- and β-linked residues. Although heparin is catalogued as a glycosaminoglycan, it is not a structural polysaccharide, but it is stored within the secretory granules of mast cells and released into the vasculature at sites of tissue injury (maybe for protection against infections, but its function is not yet clear). As a medication, it is extensively used as an **anticoagulant**.

Author: Ruben Pio

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